EFY TESTED
SANI THEO

The IoT And Touch-Based HOME AUTOMATION

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Presented here is a project based on ESP32 module to control home appliances using its touch-sensing inputs and Wi-Fi features.

ESP32 module is great for designing the Internet of Things (IoT) applications—adding touches to it makes it smarter. It is a microcontroller (MCU) designed by Espressif, mainly for IoT applications. ESP32 contains Wi-Fi, Bluetooth, inbuilt touch-sensing input pins, temperature and hall sensors on board. This makes it fit for the IoT and smart homes.

ESP32 has 10 capacitive touch-sensing general-purpose input pins. A touch-sensor system is built on a substrate that carries electrodes and relevant connections under a protective flat surface. When a user touches the surface, capacitance variation is triggered, and a binary signal is generated to indicate whether the touch is valid.

Touch-sensing pads can be arranged in different combinations (such as matrix, slider, etc), so that a larger area or more points can be detected. Touchpad-sensing process is under the control of a hardware-implemented finite-state machine (FSM), which is initiated by software or a dedicated hardware timer. In this article we learn how to handle these

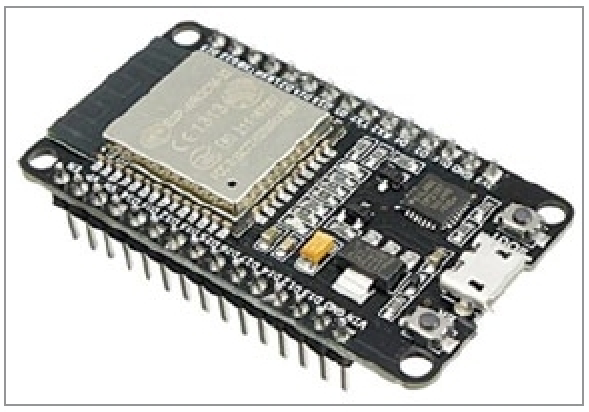

Fig. 1: ESP32 module

Fig. 4: Touchpad

Fig. 2: USB Type-C cable

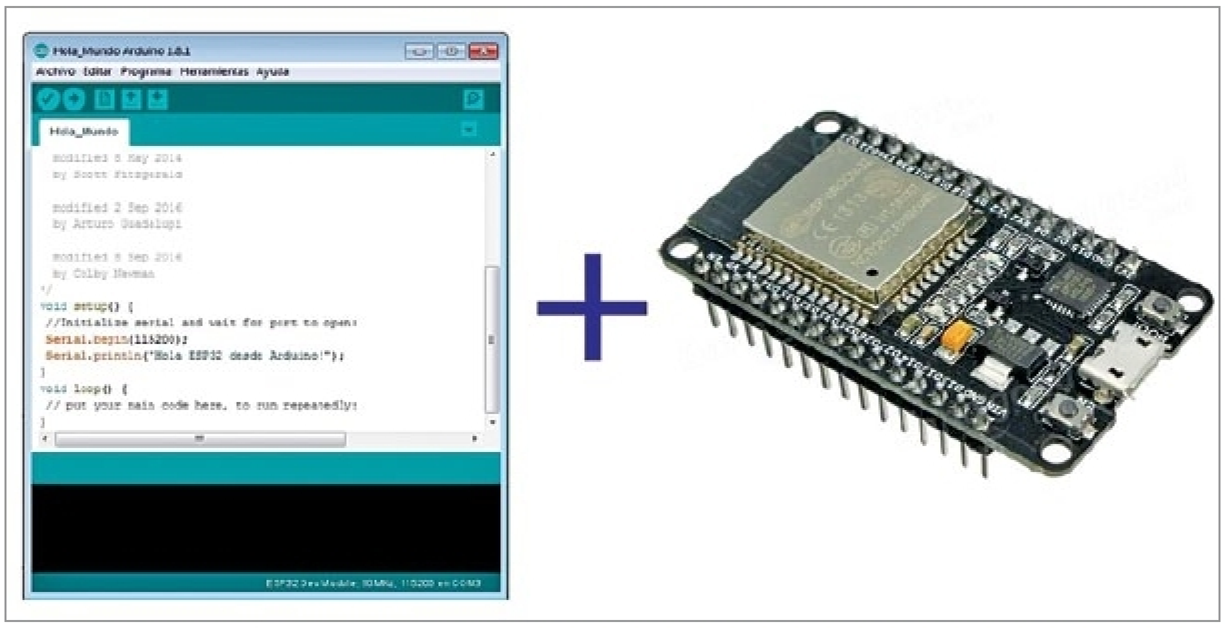

Fig. 5: Arduino IDE with ESP32 module support

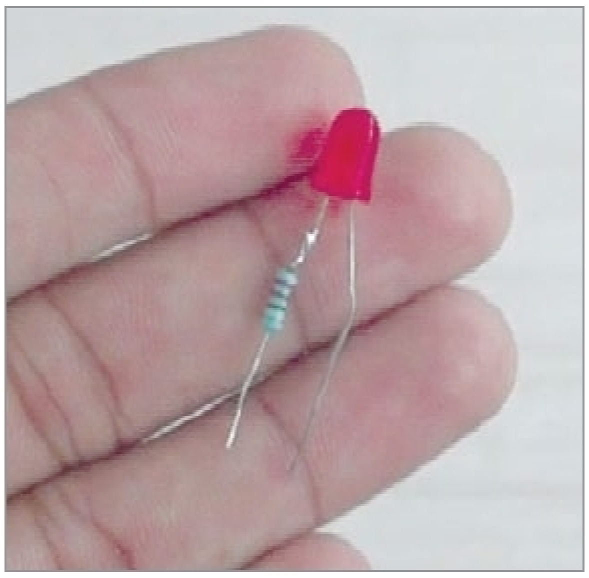

Fig. 3: LED with resistor

```
#include <WiFi.h>
const char* ssid     = "xxxx";            // Replace with your network credentials within the double quotes
const char* password = "xxxx";
WiFiServer server(80);                    //Web server port is set to 80

String header;                            // Variable to store the HTTP request
bool state=true;
String output5State = "off";              // Auxiliar variables to store the current output state
const int output5 = 5;                    // Assign an Output variable to declare GPIO pins
int s1=0;
void setup()
{
  Serial.begin(115200);
```

Fig. 6: Code declarations